Self-Supervised Masked Graph Autoencoder for Hyperspectral Anomaly Detection

Bing Tu, *Senior Member, IEEE*, Baoliang He, *Student Member, IEEE*, Yan He, *Member, IEEE*, Tao Zhou, *Student Member, IEEE*, Bo Liu, *Member, IEEE*, Jun Li, *Fellow, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*—Hyperspectral image anomaly detection faces the challenge of difficulty in annotating anomalous targets. Autoencoder(AE)-based methods are widely used due to their excellent image reconstruction capability. However, traditional grid-based image representation methods struggle to capture long-range dependencies and model non-Euclidean structures. To address these issues, this paper proposes a self-supervised Masked Graph AutoEncoder (MGAE) for hyperspectral anomaly detection. MGAE utilizes a Graph Attention Network (GAT) autoencoder to reconstruct the background of hyperspectral images and identifies anomalies by comparing the reconstructed features with the original features. Specifically, we constructs a topological graph structure of the hyperspectral image, which is then input into the GAT autoencoder for reconstruction, leveraging the multi-head attention mechanism to learn spatial and spectral features. To prevent the decoder from learning trivial solutions, we introduce a re-masking strategy that randomly masks both the input features and hidden representations during training, forcing the model to learn and reconstruct features under limited information, thereby improving detection performance. Additionally, the proposed loss function with graph Laplacian regularization (Twice Loss) minimizes variations in feature representations, leading to more consistent background reconstruction. Experimental results on several real-world hyperspectral datasets demonstrate that MGAE outperforms existing methods.**

***Index Terms*—Anomaly detection, autoencoder, graph attention network, re-masking strategy, twice loss.**

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Bing Tu, Baoliang He, Yan He, and Bo Liu are with the Institute of Optics and Electronics, the State Key Laboratory Cultivation Base of Atmospheric Optoelectronic Detection and Information Fusion, Jiangsu International Joint Laboratory on Meteorological Photonics and Optoelectronic Detection, and Jiangsu Engineering Research Center for Intelligent Optoelectronic Sensing Technology of Atmosphere, Nanjing University of Information Science and Technology, Nanjing 210044, China (e-mail: tubing@nuist.edu.cn; baoliang_he@163.com; 975861884@qq.com; bo@nuist.edu.cn).

Tao Zhou is with the School of Electronic Information and Electrical Engineering, Anhui Jianzhu University, Hefei 230601, China (e-mail: zhoutao@stu.ahjzu.edu.cn).

Jun Li is with the Faculty of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: lijuncug@cug.edu.cn).

Antonio Plaza is with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica, University of Extremadura, 10003 Cáceres, Spain (e-mail: aplaza@unex.es).

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I. Introduction

HYPERSPECTRAL imaging differs from traditional remote sensing technologies by providing simultaneous information on the radiance, geometry, and spectrum of ground objects [1], [2]. The nearly continuous spectral information provided by Hyperspectral Images (HSI) conveys a powerful ability to distinguish the subtle spectral differences between ground objects, leading to the widespread application of hyperspectral technology in various fields [3], [4]. Target detection, a key research area in HSI, can be categorized into spectral matching detection and anomaly detection based on the need for prior information. The former employs prior information of the target spectral signal for spectral matching, while the latter does not require any prior information about the targets and identifies anomalous targets based on spectral differences. In practical applications, acquiring prior spectral information of targets is often challenging [5], [6], [7]. Consequently, anomaly detection is highly practical and fundamental in HSI processing.

In recent years, numerous algorithms for hyperspectral anomaly detection (HAD) have been developed and can be typically divided into two categories:

1) Statistics-Based:

Such methods operate under the core premise that the background adheres to a specific statistical distribution, while anomalous targets deviate significantly from these statistical characteristics. The Reed-Xiaoli (RX) [8] detector is a representative statistically-based method. It quantifies the anomaly degree of a test pixel by computing the Mahalanobis distance using the global mean and covariance matrix, under the assumption that the background follows a multivariate Gaussian distribution. However, backgrounds are often complex and heterogeneous in real-world scenarios, rendering the Gaussian distribution assumption unreliable and constraining its background suppression capability. To address this limitation, the Local RX (LRX) [9] estimates local statistics within a sliding window; the Segmented RX (SRX) [10] partitions the background into homogeneous categories via clustering; and the Kernel RX (K-RX) [11] enhances separability between background and anomalies by mapping the original data into a high-dimensional feature space via kernel functions. Additionally, other statistically-based methods have been developed. For instance, Lin et al. [12] proposed a dual-dictionary construction method via two-stage complementary decisions (DDC-TSCD). Schweizer and Moura [13] introduced an anomaly detector based on Gaussian-Markov random fields (MRF). Diverging from these approaches, recent research addressing background suppression has primarily focused on low-rank and sparse matrix decomposition frameworks [14], [15]. Ren et al. [16] proposed a non-local and deep prior-based anomaly detection method (NLDPAD). Wang et al. [17] proposed an anomaly detection framework based on Tensor Low-Rank and Sparse Representation (TLRSP).

2) Non-Statistics-Based:

Unlike statistically-based approaches, non-statistical methods abandon assumptions about background distributions and instead develop alternative frameworks encompassing machine learning-based, representation learning-based, and deep learning-based techniques. Within machine learning-based approaches, these can be categorized into distance-based methods, isolation-based models and clustering-based detectors. A representative distance-based approach is the Support Vector Data Description (SVDD) anomaly detection method [18], which defines a novel detection statistic using a decision rule derived from the computation of a minimal enclosing hypersphere to discriminate anomalies. Graph-theoretic methods also fall within the distance-based category, leveraging topological relationships between pixels for anomaly identification. The Topological Anomaly Detection (TAD) algorithm stands as the first graph-based method for HAD, notably pioneering the application of graph structures in this field [19]. In TAD, the topological connection density between a test pixel and background components on the graph determines its anomaly status. Xu et al. [20] presented a systematic review of machine learning-driven HAD methods, and underscored the insight that while graph-structured models are capable of capturing spatial-spectral correlations within hyperspectral data, traditional graph-based approaches face challenges in dynamically differentiating the relative importance of neighboring nodes. Li et al. [21] developed a kernelized Isolation Forest (iForest) detector that projects the original data into a kernel space, where anomalies exhibit higher separability from the background. Recently, representation learning-based approaches have gained significant traction in HAD. Jiao et al. [22] presented a method integrating intrinsic image decomposition with background subtraction. This technique fuses a reflectance-weighted map derived from intrinsic decomposition with an anomaly probability map generated through background subtraction to produce the final detection result. Deep learning techniques have demonstrated significant impact across diverse domains, with their rapid advancement offering novel paradigms for HAD. Wang et al. [23] proposed a sliding dual-window-heuristic reconstruction network (DirectNet) for HAD, which suppresses anomalous interference during the reconstruction process through a sliding dual-window mechanism, thereby amplifying the reconstruction errors of anomalous targets. In subsequent work, Wang et al. [24] developed a pixel-shuffled undersampling blind-spot reconstruction network (PDBSNet), which imposes elevated reconstruction errors on anomalous pixels to enable detection.

In this article, we implement HAD through a self-supervised masked graph autoencoder. Our approach falls within the non-statistical methodology category. Although grounded in graph theory, it should not be characterized solely as distance-based since its core mechanism relies on autoencoder (AE) architecture. Initially proposed by Baldi in 2006 [25], AE constitutes a powerful unsupervised learning model capable of autonomously learning intrinsic data structures and patterns without manual feature engineering. Subsequent variants – including deep autoencoders and convolutional autoencoders – have significantly enhanced representation capacity through multi-layered neural frameworks. These advancements have established AE's efficacy in image processing, data compression, and feature learning tasks, demonstrating unique advantages for HAD applications in complex scenarios. Emoto and Matsuoka [26] integrated tensor robust principal component analysis (TRPCA) with autoencoder adversarial networks to learn nonlinear low-dimensional representations of spectral features in background regions. Wang et al. [27]proposed a frequency-to-spectral mapping generative adversarial network (GAN) for HAD. This network maps raw spectra to the fractional Fourier domain, transforms the anomaly identification task into a mapping task, and leverages reconstruction errors to distinguish between backgrounds and anomalies. Wu et al. [28] proposed a lightweight unsupervised cross-image HAD method (VJDNet), which utilizes variational autoencoders (VAE) to learn global and local discriminative features of anomalies. Zhao et al. [29] proposed two cascaded autoencoders with adaptive pixel-level attention modules (CAPNet) for HAD. Through cascaded autoencoders, discriminative high-level semantic features can be effectively extracted, enhancing the difference between anomalies and the background. The core advantage of AE lies in its precise reconstruction capability for normal data. For data points deviating from a normal distribution, AE often fails to achieve effective reconstruction, thereby producing significant reconstruction errors.

In this article, we employ a re-masking strategy to prevent the decoder from learning trivial mapping relationships. As a fundamental technique in computer science, masking selectively processes data through binary patterns – enabling suppression of irrelevant information and enhancement of salient features. Background suppression was primarily achieved indirectly through statistical methods prior to 2010. For instance, the Reed-Xiaoli (RX) algorithm implicitly masks distribution-conforming pixels by assuming Gaussian background distributions and leveraging covariance matrices, yet remains constrained by rigid statistical assumptions and linear modeling capabilities. With the rise of deep learning, masking strategies have gained increased attention. Sun et al. [30] proposed Contrastive Self-Supervised Background Reconstruction (CSSBR) for HAD, which constructs a self-supervised pretraining model through pixel and patch-level masking combined with dual-attention networks. This framework learns generalized background representations without requiring annotated samples. Wang et al. [31]

introduced a fourier conditional masking with Mixed Attention (FCMMA) method, where conditional masks (CMASK) suppress high-frequency anomaly components while preserving low-frequency background information in the frequency domain, thereby optimizing detection performance. In their subsequent research, Wang et al. [32]further proposed a non-local and local feature-coupled self-supervised network (NL2Net) for HAD. This network incorporates masked convolution to strengthen the focus on background features, thereby effectively addressing the non-local self-similarity issue in HSI. Su et al. [33] utilized latent binary masks to separate anomalies from backgrounds, reconstructing backgrounds while suppressing anomalies via separation loss functions. The recently proposed hyperspectral foundation model HyperSIGMA [34] pinpoints the core bottleneck of traditional models to be their inadequate general feature representation capability, most methods are limited to specific scenarios (e.g., single homogeneous backgrounds) and exhibit significant performance degradation in complex heterogeneous scenes, primarily due to insufficient modeling of feature correlations. While HyperSIGMA enables robust feature extraction in complex scenes via multimodal fusion and self-supervised masked learning, its design as a general foundation model means it lacks optimization for the balance between background reconstruction and anomaly discrimination. Notably, this latter shortcoming is critical to background consistency modeling in HAD. Therefore, within an unsupervised framework, developing an approach to integrate graph structures with masked learning for accurate background reconstruction and efficient anomaly discrimination in complex scenes remains a pressing open challenge in the field.

To address these issues, this study proposes a self-supervised masked graph autoencoder (MGAE) for HAD. This method utilizes a graph attention network (GAT) to construct an autoencoder. By reconstructing the hyperspectral image background and comparing the reconstructed features with the original features to identify anomalies, it effectively captures the spatial-spectral dependencies and non-Euclidean structure of the hyperspectral data. Specifically, the MGAE first constructs the topological graph structure of the hyperspectral image and feeds it into the GAT autoencoder for reconstruction. Leveraging a multi-head attention mechanism [35], [36], [37], the MGAE learns both spatial and spectral features, overcoming the limitations of traditional grid representations, which struggle to model long-range dependencies. Secondly, a re-masking strategy is introduced, as shown in Fig. 1. During training, the input features and hidden representations are randomly masked, forcing the model to learn and reconstruct features with limited information [38], [39], avoiding the issue noted in [20] that the decoder of self-supervised autoencoders is prone to trivial solutions and suffers from reconstruction error averaging. Furthermore, a loss function (Twice Loss) with graph Laplacian regularization is designed to achieve more consistent background reconstruction by minimizing changes in feature representations. At the same time, it imposes minimal constraints on anomalous regions to prevent them from being reconstructed, achieving a balance between

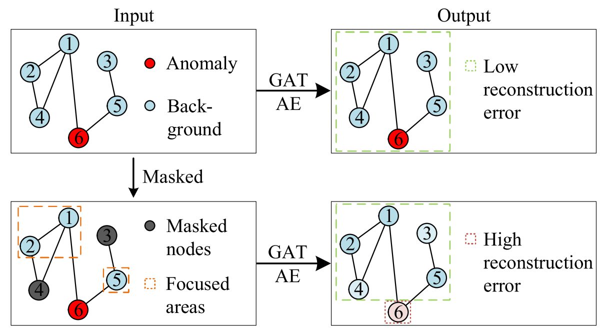

Fig. 1. Comparison between original input and masked input. The original input may lead to simple identity mapping, causing the model to ignore anomalies due to uniformly low reconstruction error. Masking forces the model to reconstruct the masked areas based on unmasked nodes, focusing more on learning background features.

local consistency and global information preservation. The main contributions of our work can be summarized as follows:

- By integrating GAT into the anomaly detection task, its multi-head attention mechanism can dynamically assign weights to neighborhood nodes, flexibly modeling complex relationships between pixels, enhancing sensitivity to key areas and suppressing noise interference. This effectively captures the spatial-spectral dependency of hyperspectral data, and improves the detection capability of anomalies of different scales and shapes.
- By performing double random masking on the input features and hidden representations, the model is forced to learn and reconstruct features under limited information, avoiding the problem of reconstruction error averaging caused by the decoder learning a simple mapping relationship. This significantly improves the model's generalization ability and adaptability to data incompleteness, and demonstrates more stable detection performance on real datasets.
- A Twice Loss is proposed that combines reconstruction loss and graph Laplacian smoothing loss. It achieves local consistency of background reconstruction and global information preservation by minimizing feature representation changes, while imposing minimum constraints on abnormal areas to prevent them from being reconstructed, thus balancing background modeling accuracy and abnormality recognition sensitivity.

The paper is organized as follows: Section II introduces the KNN algorithm for graph construction, the GAT model and masking strategy. Section III details the proposed method, including the GAT-based autoencoder, the re-masking strategy, and the Twice Loss function. Section IV demonstrates the method's effectiveness through experiments on real HSI. Finally, Section V summarizes the work and discusses future research directions.

II. Related Work

Graph structures are a powerful form of data representation capable of flexibly depicting irregular and non-Euclidean spatial relationships. However, due to the variable node degrees

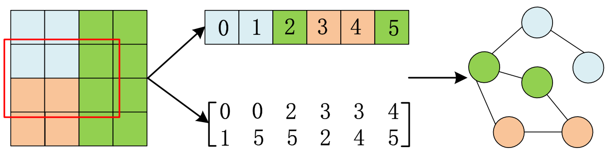

Fig. 2. Graph construction from image, with different node and edge types and related information.

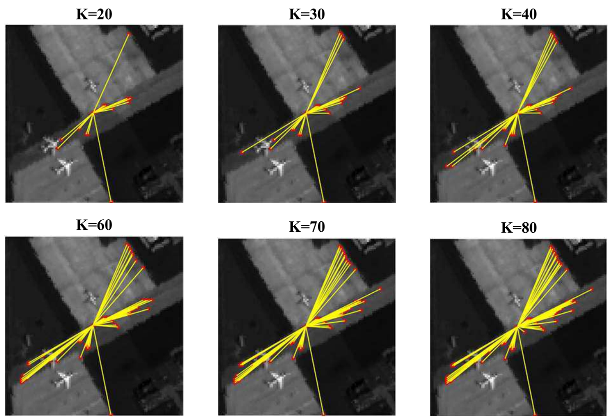

Fig. 3. Graph constructed with various K values for the central pixel in Airport-I image.

and the unordered nature of neighboring nodes in graphs, traditional deep learning techniques are not directly applicable to graph data. Before the advent of GCN, feature extraction in graphs primarily relied on manually designed features or rule-based methods. The emergence of GCN ingeniously combined graph theory with Convolutional Neural Network (CNN), utilizing spectral and spatial convolution techniques to efficiently model graph data. The spectral convolution approach is based on the graph Fourier transform, which processes graph signals by converting them to the frequency domain. In contrast, spatial convolution updates node information through the aggregation of neighborhood information and feature transformation. In this paper, we construct the graph structure using the K-Nearest Neighbors (KNN) algorithm [40], [41], [42] and employ the GAT for learning node features. In this section, we will introduce the foundational knowledge used in this paper.

A. Preliminaries

The graph structure is composed of a finite and non-empty set of vertices and a set of edges between vertices, typically represented as $G = (V, E)$. Here, G denotes a graph, V is the set of vertices of graph G, and E is the set of edges of graph G. Unlike image data that relies on a fixed grid structure, graph data can more flexibly represent the complex spatial and spectral relationships in hyperspectral data. By establishing adjacency relationships, the graph structure effectively captures both local and global similarities between pixels, making it particularly suitable for dealing with sparse features and non-uniform distributions in hyperspectral data. Therefore, in this paper, we choose to construct graph data to perform anomaly detection tasks for HSI.To convert hyperspectral image data into a graph structure, the edge set E is constructed via the K-nearest neighbor algorithm. The K-Nearest Neighbors (KNN) algorithm is a method used to construct edges based on spectral characteristic similarity, it utilizes a defined distance metric (typically Euclidean distance) to find the k-nearest neighbors for each data point, i.e., each pixel in the image. In hyperspectral images, each pixel contains spectral data from multiple bands, forming a high-dimensional feature space. KNN evaluates and identifies the most similar neighbors in this feature space. By using each pixel's spectral data as a feature vector and calculating the Euclidean distances between every pair of pixels, the k closest neighbors are identified. Edges are then constructed in the graph based on the connections between each pixel and its k-nearest neighbors, thus indicating spectral similarity. The formula can be expressed as:

$$d_{\text{spec}}(p_i, p_j) = \sqrt{\sum_{b}^{b=1} (x_{ib} - x_{jb})^2} \tag{1}$$

B. Graph Attention Networks

The rise of graph neural networks (GNN) has provided a new paradigm for modeling the structure of hyperspectral data. Graph Attention Networks (GAT), in particular, overcome the limitations of traditional graph convolutional networks (GCN) by introducing an attention mechanism, becoming a valuable tool for processing non-Euclidean data. Early graph-based HAD methods attempted to capture inter-pixel relationships, but suffered from significant limitations. The emergence of GCN has promoted the integration of graph structures with deep learning. GCN aggregates information through weighted averaging of neighborhood features, has shown potential in modeling hyperspectral spatial-spectral relationships. However, it relies on the homogeneous graph assumption, assuming that all nodes and edges are of the same type, making it difficult to handle the heterogeneous structure prevalent in hyperspectral data. Feature aggregation relies on predefined fixed weights, failing to dynamically capture complex inter-node relationships and making them susceptible to noise. Simple weighted averaging operations struggle to distinguish the importance of neighboring nodes, diluting key features sensitive to anomalies. Subsequent improvements, such as dynamic graph models and multi-scale feature extraction, have alleviated these issues to some extent, but remain limited in their ability to model node relationships, and their performance on sparse or heterogeneous graphs remains suboptimal.

In contrast, GAT dynamically assigns node weights through an attention mechanism, providing a more flexible solution for HAD.It eliminates the need to rely on graph homogeneity assumptions and automatically distinguishes the importance of neighboring nodes by learning attention coefficients, effectively handling heterogeneous scenes with mixed feature types in hyperspectral data. The multi-head attention mechanism extracts node features from multiple perspectives, enhancing the ability to capture anomalies of varying scales and spectral characteristics. Dynamic adjustment of attention weights enables the model to adaptively suppress interference from noisy nodes, improving sensitivity to subtle anomalies [43], [44], [45].

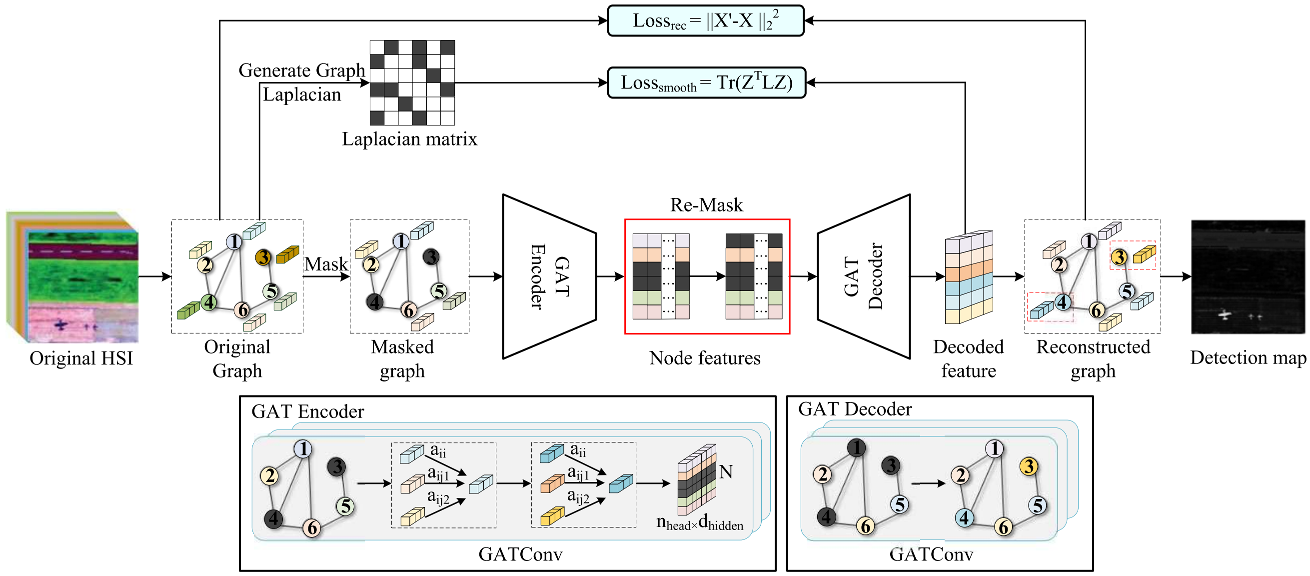

Fig. 4. The proposed MGAE-based HAD flow chart. First, the input hyperspectral image is constructed as a spatial topological graph using the KNN algorithm. Subsequently, a preliminary masking operation is performed on the node features, and the masked graph data is input into the GAT encoder. At the encoder output layer, a re-masking strategy is used to mask the hidden layer representations twice. The processed features are then passed through the GAT decoder to reconstruct the original spectral features. Finally, a dual loss function is used to jointly optimize the reconstruction error and the graph structure smoothness constraint to output the anomaly detection result graph.

The MGAE proposed in this study further expands the application of GAT in HAD. Unlike methods that rely solely on GAT for feature aggregation, MGAE combines GAT with an autoencoder architecture to reconstruct background through an "encode-decode" process. A re-masking strategy is introduced to force the model to learn more robust feature representations, avoiding the "trivial mapping" problem that traditional GAT often falls prey to. Furthermore, by constraining feature smoothness through a graph Laplacian regularization loss, the topological relationships captured by GAT are more consistent with the spatial continuity of hyperspectral data, further improving the separability of background and anomalies. This design not only retains GAT's ability to model complex relationships but also enhances the stability and accuracy of detection through self-supervised learning and regularization mechanisms.

C. Masking Strategy

Masking is a technique that precisely controls specific bits of target data through binary bitwise operations. Its core approach involves extracting, modifying, filtering, or flipping specific bits in the target data using a binary sequence of the same length as the target data through bitwise operations such as AND, OR, and XOR. It is widely used in tasks such as classification, anomaly detection, and background suppression to achieve refined control over binary data.

This study introduces a re-masking strategy that applies double AND operations to deeply couple the masking process with graph topology. This shifts the model's background learning from "individual pixel features" to "neighborhood correlation patterns". Ultimately, anomalous nodes exhibit elevated reconstruction errors due to violations of background correlation patterns, enabling effective discrimination between anomalies and background. This design not only ensures the core capability of the mask operation to "selectively retain key information" but also breaks through the traditional mask's dependence on linear relationships and static scenes through graph structure guidance, and significantly improves the robustness of HAD.

III. Proposed Methodology

In this section, we initially outline our approach. Subsequently, we elaborate on the implementation specifics of each element of our method.

A. GAT-Based Autoencoder

Our proposed anomaly detection method consists of three main steps: topology graph generation, re-masking operation, and graph autoencoder reconstruction based on GAT. The overall architecture of MGAE is shown in Fig. 4.

First, we load the three-dimensional HSI as a node feature matrix x, where each pixel in the image is treated as a node. To capture the spatial relationships between pixels, adjacency information is constructed using the KNN algorithm. This step creates a topology graph, which represents the connectivity between neighboring pixels based on their feature similarities. The adjacency matrix reflects the inherent spatial and spectral dependencies in hyperspectral data, which is crucial for capturing the correlations between different spectral bands and pixels in the image. It can be expressed with the following formula:

$$E = \mathrm{KNN}(x, x, k) \tag{2}$$

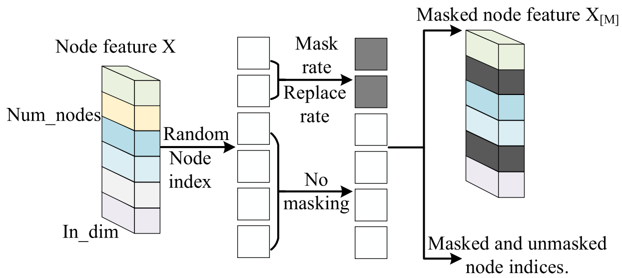

Fig. 5. Random masking operation in the re-masking process.

where E is the set of edges and k is the parameter in the KNN algorithm that specifies the number of nearest neighbor nodes each node should connect to.

Subsequently, an initial masking operation is applied to the node feature matrix x, where a random subset of features is set to zero. The masked feature matrix, along with the adjacency matrix, is then fed into a GAT-based encoder consisting of multiple GATConv layers. The encoder learns to capture spatial and spectral dependencies through a multi-head attention mechanism. Each attention head independently computes the attention coefficients a_{ij}, representing the relationship strength between node i and node j. The encoder aggregates the node features based on these attention coefficients, producing a new node feature representation. This process retains inter-node relationships while combining spatial and spectral information. The encoder outputs the latent representation *rep*, with dimensions $n_{head} \times d_{hidden}$, where n_{head} is the number of attention heads and d_{hidden} is the hidden feature dimension. This latent representation captures node relationships and feature information, serving as the basis for the subsequent reconstruction process. A second masking operation is applied to the encoder output as part of our proposed re-masking strategy, which helps the model learn more robust feature representations by reconstructing the missing parts of the data.

$$x_{masked} = x \odot M_1 \tag{3}$$

$$rep = \text{Encoder}(x_{masked}, E) \tag{4}$$

$$rep_{remasked} = rep \odot M_2 \tag{5}$$

where the symbol $\odot$ denotes element-wise multiplication, and M_1 and M_2 are the initial mask matrix and the re-mask matrix, respectively. The masked positions are set to 0, while all other positions are set to 1.

The re-masked *rep* is then passed to the GAT-based decoder, which reconstructs the node feature matrix $\hat{x}$. The decoder attempts to recover the original features of the nodes based on the encoded and re-masked representation. The output of this stage is an approximation of the original hyperspectral data, which is compared to the actual data during training to compute the reconstruction loss.

$$\hat{x} = \text{Decoder}(rep_{remasked}, E) \tag{6}$$

B. Re-Masking Strategy

The proposed re-masking strategy is applied at the data input stage and the encoder output stage. The masking method is shown in the Fig.5. Its design principle is as follows:

1) Masked Feature Reconstruction: When the mask dimension exceeds the input dimension, ordinary autoencoders tend to learn meaningless "identity functions", making the encoded representation useless. This problem is particularly prominent in graph data with small node feature dimensions, and existing graph autoencoders often ignore this problem. Inspired by denoising autoencoders which avoid trivial solutions by deliberately corrupting the input, we use masked autoencoders as the basic framework of MGAE. In our method, a portion of nodes is randomly selected and their features are replaced with zero. The resulting masked feature matrix defines the characteristics of each node as follows:

$$\tilde{x}_i = \begin{cases} 0, & v_i \in \widetilde{V} \\ x_i, & v_i \notin \widetilde{V} \end{cases} \tag{7}$$

The goal of MGAE is to reconstruct masked node features using partially observed node signals and an input adjacency matrix. By randomly selecting a subset of nodes, setting their features to zero, and performing uniform random sampling without replacement, the mask distribution is balanced (preventing all nodes' neighbors from being completely masked or visible). Finally, the masked node features are reconstructed using the partially observed node signals and the input adjacency matrix. In graph autoencoders, nodes rely on neighbor information to optimize their features, and this sampling strategy effectively maintains this balance.

2) GAE Decoder With Re-Mask Decoding: The decoder needs to map the latent code back to the input features, and its design is related to the semantic complexity of the input. For graph data, traditional graph autoencoders often use non-neural network decoders or basic multi-layer perceptrons, which has limited expressive power. This results in the latent code being highly similar to the input features, making it difficult to form a compressed and information-rich representation. To address this, MGAE uses a more expressive single-layer graph attention network as the decoder. This reconstructs features based on node neighborhood information rather than relying solely on the nodes themselves, enabling the encoder to learn more abstract latent representations.

To further advance compressed representation learning, we introduce a re-masking technique: the masked node indices are reset to zero again at the latent code level to simulate additional noise. This forces the masked nodes to reconstruct their input features using the latent representations of their unmasked neighbors. The re-masked latent code is defined as:

$$\tilde{h}_i = \begin{cases} 0, & h_i \in \widetilde{V} \\ h_i, & h_i \notin \widetilde{V} \end{cases} \tag{8}$$

C. Twice Loss

Our framework employs a dual loss function named Twice Loss, as illustrated in the Fig. 6. Subsequently, we will elaborate on the conceptual design of the loss function.

1) Reconstruction Loss: We use the squared L_2 norm to calculate the loss, which imposes a greater penalty on larger errors. This means that if some pixels have larger differences,

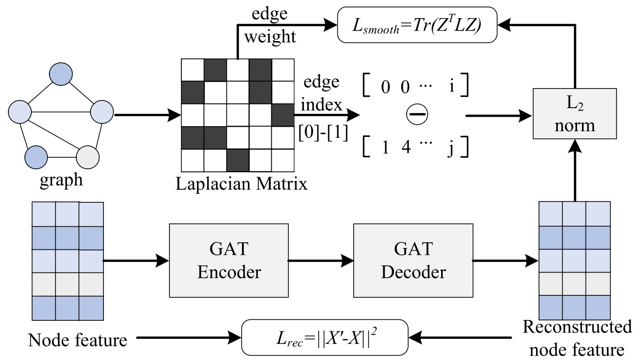

Fig. 6. The implementation of the proposed twice loss.

their contribution to the loss will also be greater, and the model will work harder to reduce these differences. This characteristic makes the squared L_2 norm suitable for image reconstruction tasks that are sensitive to global errors.

$$\mathcal{L}_{rec} = \|X' - X\|_2^2 \tag{9}$$

where $\mathcal{L}_{rec}$ is the reconstruction loss, which measures the difference between the input X and the reconstructed output X'. The goal is to minimize this loss, making the reconstructed X' as close as possible to the input. X' denotes the reconstructed output, i.e., the result obtained after the reconstruction network model processes the input, representing the hyperspectral image data that the model tries to restore; X represents the input data, i.e., the original image data, which serves as the "ground truth" for the model during reconstruction. We hope that X' can be as close as possible to X.

2) Smooth Loss: Although the traditional reconstruction loss effectively measures the difference between the input and output images, it treats all pixels 'equally' without considering the local structural relationships between pixels (features). This approach may lead the model to reconstruct all areas (including anomalous regions) to the same extent in HAD, thereby affecting the detection performance. To further enhance the network's effectiveness in reconstructing the background of hyperspectral images and to utilize the local structural information in the data, we introduce a quadratic loss based on graph Laplacian regularization.

$$\mathcal{L}_{smooth} = \text{abs}\left[\text{Tr}(Z^T LZ)\right] \cdot 0.00001 \tag{10}$$

where Z represents the feature representation output by the model, L is the graph Laplacian matrix, and $\mathcal{L}_{smooth}$ is the smoothness loss, also referred to as the second loss. The graph Laplacian matrix L, defined through the graph's adjacency matrix and degree matrix, captures the local relationships between pixels or features in the data. The primary purpose of regularization is to guide the model to maintain consistency in features during the training process, especially in the background areas, ensuring that the reconstruction between adjacent pixels remains smooth and consistent. The factor of 0.00001 is multiplied to balance the magnitudes of the two terms, ensuring the effectiveness of the loss function.

By minimizing the smoothness loss, the model reduces the differences among adjacent feature points in high-dimensional space. This means that the background areas will exhibit a smoother reconstruction effect, while the anomalous areas, due to weaker connectivity in the graph structure, are less constrained by the model in their reconstruction, which helps enhance the identification and detection of anomalous areas. The smoothness loss not only improves the accuracy of background reconstruction but also enhances anomaly detection sensitivity by reducing over-reconstruction in anomalous areas.

3) Twice Loss: In summary, we define the loss function $\mathcal{L}$ for MGAE as follows:

$$\begin{aligned}\mathcal{L} &= \alpha \cdot \mathcal{L}_{\text{rec}} + (1-\alpha) \cdot \mathcal{L}_{\text{smooth}} \\ &= \alpha \cdot \|X' - X\|_2^2 + (1-\alpha) \cdot \text{abs}\left[\text{Tr}(Z^T LZ)\right] \cdot 0.00001 \end{aligned} \tag{11}$$

where $\mathcal{L}_{rec}$ is the reconstruction loss, $\mathcal{L}_{smooth}$ is the smoothness loss, and α is a hyperparameter. α controls the strength of the graph Laplacian smoothness loss. By adjusting α, we can flexibly balance the trade-off between smoothness in background reconstruction and sensitivity in anomaly detection. When α is large, the model emphasizes local smoothness more in the reconstruction process, which is suitable for optimizing background areas; when α is small, the model handles feature changes in anomalous areas more flexibly, avoiding excessive smoothing of anomalous areas.

D. Anomaly Scores

Anomalies are measured by the difference (*diff*) between the reconstructed and original node feature matrices. The L_2 norm and Mahalanobis distance are computed from this difference and combined with weights to produce a comprehensive anomaly score, which serves as the final detection result. Their definitions are as follows:

1) L_2 Norm: Measures the Euclidean distance between the original data and the reconstructed data. The calculation formula for scores_l2 is shown as follows:

$$\text{diff} = x - \hat{x} \tag{12}$$

$$\text{scores\_l2} = \sum_{j=1}^{d} (\text{diff}_{ij})^2 \tag{13}$$

where *diff* represents the difference between the original feature matrix x and the reconstructed feature matrix $\hat{x}$. x is the original feature matrix, $\hat{x}$ is the reconstructed feature matrix, i is the node index, j is the feature dimension, and d is the total number of features.

2) Mahalanobis Distance: Incorporating the covariance matrix, it measures the deviation of node features from the global distribution. The calculation formula for scores_ma is shown as follows:

$$\Sigma = \text{Cov}(\text{diff}) \tag{14}$$

$$\Sigma^{-1} = (\text{Cov}(\text{diff}))^{-1} \tag{15}$$

$$\mu = \frac{1}{n} \sum_{i=1}^{n} \text{diff}_i \tag{16}$$

$$\text{scores\_ma} = \text{diag}\left((\text{diff} - \mu) \cdot \Sigma^{-1} \cdot (\text{diff} - \mu)^\top\right) \tag{17}$$

where Σ represents the covariance matrix of the differences *diff*, and Σ^{-1} is its inverse, normalizing distances by accounting for feature correlations. μ is the mean vector of *diff*, and n

Algorithm 1 Proposed MGAE for Hyperspectral Anomaly Detection

Require: HSI image $\mathbf{X} \in \mathbb{R}^{H \times W \times B}$, Parameters: k (KNN neighbors), α (loss weight).

Ensure: Anomaly detection DetectMap $\mathbf{D} \in \mathbb{R}^{H \times W}$, Mean AUC score across random seeds

1: **Stage1:Training Phase**

2: Obtain the graph structure input $\mathbf{X}_{flat}$ converted from HSI by using the KNN algorithm with k value.

3: Generate binary *re-masks M1, M2* , mask before encoding: $\mathbf{X}_{masked}$ = $\mathbf{X}_{flat} \odot M1$; Encode: Z = *Enc*($\mathbf{X}_{msaked}$, E), *re-masked:* $\mathbf{X}_{re-msaked} = Z \odot M2$.

4: Reconstruct full input: $\hat{X}_{\text{val}}$ = *Dec(Enc(*$\mathbf{X}_{flat}$, *E) E)*; Compute anomaly scores from $\hat{X}_{\text{val}} - \mathbf{X}_{flat}$.

5: Training MAGE by Twice Loss Eq. (11).

6: Track AUC on validation labels, save $\mathbf{Model}_{best}$ (highest AUC).

7: **Stage2:Detection Phase**

8: Use a specific k value to perform graph structure transformation operations to obtain $\mathbf{X}_{flat}$.

9: By inputting $\mathbf{X}_{flat}$ into the trained $\mathbf{Model}_{best}$, the Decoded feature is obtained after GAT decoding.

10: Recover the reconstructed HSI through inverse operations $\mathbf{D} \in \mathbb{R}^{H \times W}$.

11: Calculate the degree of anomaly of each pixel in X by Eq. (18).

is the number of nodes. Finally, diag(·) extracts the diagonal elements, yielding the Mahalanobis distance scores for each node.

We use λ to combine scores_l2 and scores_ma with weighting, resulting in a method for calculating anomaly scores. The formula is as follows:

$$\text{Anomaly Scores} = \lambda \cdot \text{scores\_l2} + (1 - \lambda) \cdot \text{scores\_ma}. \quad (18)$$

During training, λ is set to 0.85 to focus on the model's reconstruction error. During testing, λ is set to 0.5, equally weighting the L_2 norm and Mahalanobis distance. The L_2 norm captures pointwise feature deviations, while the Mahalanobis distance detects global distribution anomalies, resulting in a more comprehensive anomaly score for detection.

IV. Experiments

In order to verify the detection performance of our proposed method, a series of contrast tests are set up in this section.

A. Experimental Datasets

In this article, the proposed method is evaluated on six real hyperspectral data sets captured at different scenes, which are listed as follows.

1) *BEACH-I*: The first hyperspectral data set was acquired by the Raster Object-oriented Sensor Image System (ROSIS-03) sensor covering the Pavia. This scene covers area of 100×100 pixels with 224 spectral channels in wavelengths ranging from 370 to 2510 nm. In the experiments, a total of 102 bands are used after removing the water-absorption bands. The spatial resolution is approximately 1.3m.
2) *BEACH-II*: The second hyperspectral data set was acquired by the AVIRIS sensor covering the Bay Chamoagne. This scene covers area of 100×100 pixels with 188 spectral channels in wavelengths ranging from 370 to 2510 nm. The spatial resolution is approximately 4.4m.
3) *HYDICE*: The third hyperspectral dataset, acquired by the AVIRIS sensor, covers an urban area in California. The scene has a pixel size of 100 × 100 pixels, an unknown wavelength range, 205 spectral channels, and an anomaly ratio of 0.91%. The spatial resolution is 1 meter.
4) *Gainesville*: The fourth hyperspectral dataset, acquired by the AVIRIS sensor, covers the Gainesville area. This scene has a pixel size of 100 × 100 pixels, an unknown wavelength range, 191 spectral channels, and 0.52% anomaly ratio. The spatial resolution is 3.5 meters.
5) *Gulfport*: The fifth hyperspectral dataset, acquired by the AVIRIS sensor, covers the Gulfport area. This scene has a pixel size of 100 × 100 pixels, an unknown wavelength range, 191 spectral channels, and an anomaly ratio of 0.60%. The spatial resolution is 3.4 meters.
6) *Los Angeles*: The sixth hyperspectral dataset, acquired by the AVIRIS sensor, covers the Los Angeles area. This scene has a pixel size of 100 × 100 pixels, an unknown wavelength range, 205 spectral channels, and an anomaly ratio of 0.87%. The spatial resolution is 7.1 meters. The image and ground truth of this dataset are shown respectively in Fig. 7 (a) and (b).

B. Comparison Algorithms and Evaluation Criteria

1) Comparison Method: This paper compares the proposed MGAE algorithm with six existing HAD algorithms, including RX [8], LRX [9], LSMAD [46], FrFE [47], BockNet [48], PUNNet [49] and PDBSNet [24]. The RX algorithm is a classic method based on statistical modeling, while LRX is a localized version of the RX algorithm. LSMAD is a representative method based on low-rank and sparse matrix decomposition. The experimental parameter initial rank l0 is set to a value matching 0.6 times the number of bands in the HSI, and the initial number of mixed Gaussian noise K is set to 4. FrFE is a complex method based on the fractional Fourier transform and entropy principles that enhances HAD by optimizing signal enhancement and noise suppression. BockNet is a self-supervised blind-block reconstruction network with a guard window. It uses the blind-block (inner window) to protect central pixels and only leverages outer window information to predict central pixels, thus increasing the reconstruction error of anomalous pixels. PUNNet is an anomaly detection method using a global feature injection blind-spot network. PDBSNet employs a pixel-reorganized downsampling blind-spot reconstruction mechanism, where the center pixel is masked as the blind spot and its spectral information is reconstructed from the surrounding neighborhood. RX, LRX, and FrFE are non-parametric techniques. For BockNet, we employ the parameters reported in its original

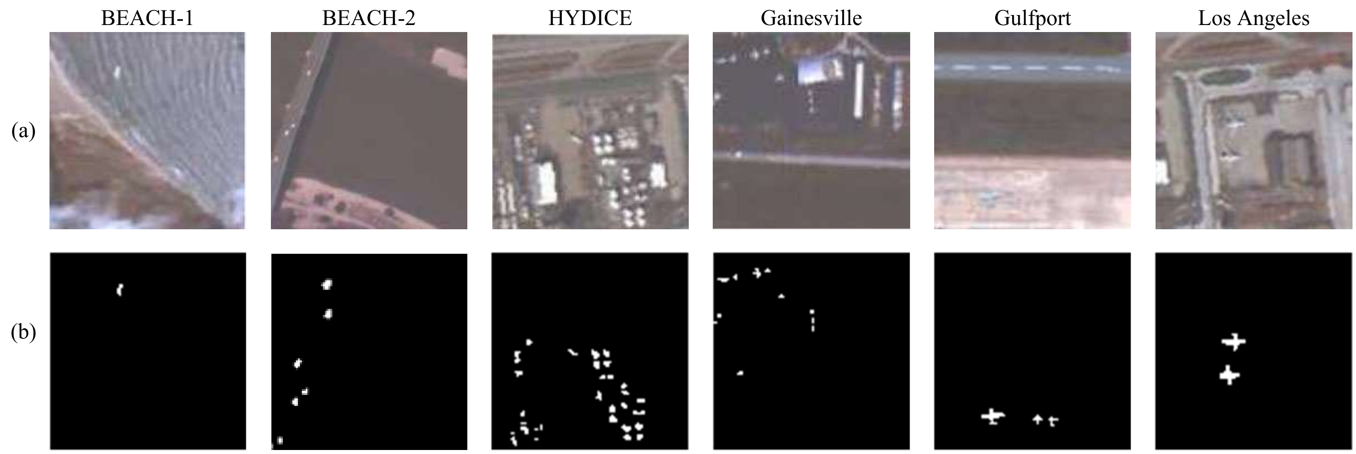

Fig. 7. Pseudo-color images and ground-truth maps of the six selected datasets. (a) Pseudo-color images. (b) Ground-truth maps.

TABLE I
AUC VALUES OF THE COMPARISON ALGORITHMS AND OUR PROPOSED ALGORITHM ON THE SIX SELECTED DATASETS

Dataset	RX	LRX	LAMAD	FrFE	BockNet	PUNNet	PDBSNet	MGAE
BEACH-I	0.9998	**0.9999**	0.9994	0.9995	0.9992	0.9993	0.9987	**0.9999**
BEACH-II	0.9886	0.9239	0.9828	0.9908	0.9932	0.9901	0.9917	**0.9946**
HYDICE	0.9691	0.9284	0.9612	0.9717	0.9493	0.9612	0.9677	**0.9741**
Gainesville	0.9512	0.8999	0.9625	0.9612	0.9896	**0.9925**	0.9858	0.9913
Gulfport	0.9525	0.7202	0.9868	0.9849	0.9912	0.9857	0.9898	**0.9923**
Los Angeles	0.8403	0.8416	0.9216	0.9153	0.9457	0.9312	0.9388	**0.9833**
Average	0.9501	0.8857	0.9691	0.9701	0.9780	0.9768	0.957	**0.9892**

paper. PUNNet utilizes kernel sizes of (1, 4, 5, 5, 5, 4). PDBSNet is configured with kernel sizes ((4, 4), (5, 5), (5, 5), (5, 5), (4, 4), (5, 5)). Our algorithm has two parameters, k and α. In the comparative experiments, we used the optimal parameters selected through experimentation. The specific parameter analysis will be discussed in detail in the later section on parameter analysis.

2) Evaluation Criteria: Based on the real data provided in Fig. 7, we can deeply evaluate the performance of each detector with the help of 3D_ROC curves and 2_ROC curves. By analyzing the three corresponding 2D_ROC curves – the relationship between detection rate P_d and false alarm rate P_f, detection rate P_d and threshold τ, and false alarm rate P_f and threshold τ – we can clearly understand the performance differences of different anomaly detectors in the relationship among P_d, P_f, and threshold τ.

C. Comparison Experiments and Detection Performance

We conducted comparison experiments of our algorithm with others on six datasets, and the detection results are shown in Fig. 8. The AUC values of each algorithm are listed in Table I. We will analyze the strengths and weaknesses of each algorithm and their detection performance.

1) 2D_ROC: The Fig. 9 illustrates the True Positive Rate (TPR) performance curves of multiple anomaly detection algorithms across six hyperspectral datasets versus the decision threshold τ. The vertical axis quantifies the algorithms' anomaly detection capability (TPR), while the horizontal axis determines the anomaly/background classification boundary. The curves exhibit a general trend of increasing TPR with decreasing threshold, but reveal distinct performance characteristics across algorithms and datasets.The MGAE algorithm consistently exhibits steeper performance curves across all datasets, rapidly approaching a TPR of 1 at low thresholds ($\tau < 0.2$) – for instance, achieving TPR > 0.95 on the BEACH-1 dataset. This behavior indicates exceptional sensitivity to subtle anomalies and significant robustness to threshold selection. In contrast, traditional algorithms (e.g., RX, LRX) display flatter, lower curves, requiring substantially higher thresholds ($\tau > 0.4$) to approach MGAE's TPR performance, demonstrating pronounced parameter dependence.MGAE's performance advantage is most distinct in homogeneous scenes like BEACH. Under the complex background of HYDIC, MGAE maintains a clear lead over methods such as BockNet and PUNNet. On the Gainesville and Gulfport datasets, while PUNNet approaches MGAE's performance within mid-range thresholds ($0.2 \sim 0.5$), it fails to match MGAE's effectiveness at low thresholds ($\tau < 0.2$). For the spectrally confounded Los Angeles scene, all algorithms exhibit marginally reduced peak TPR. However, MGAE sustains its superior performance,

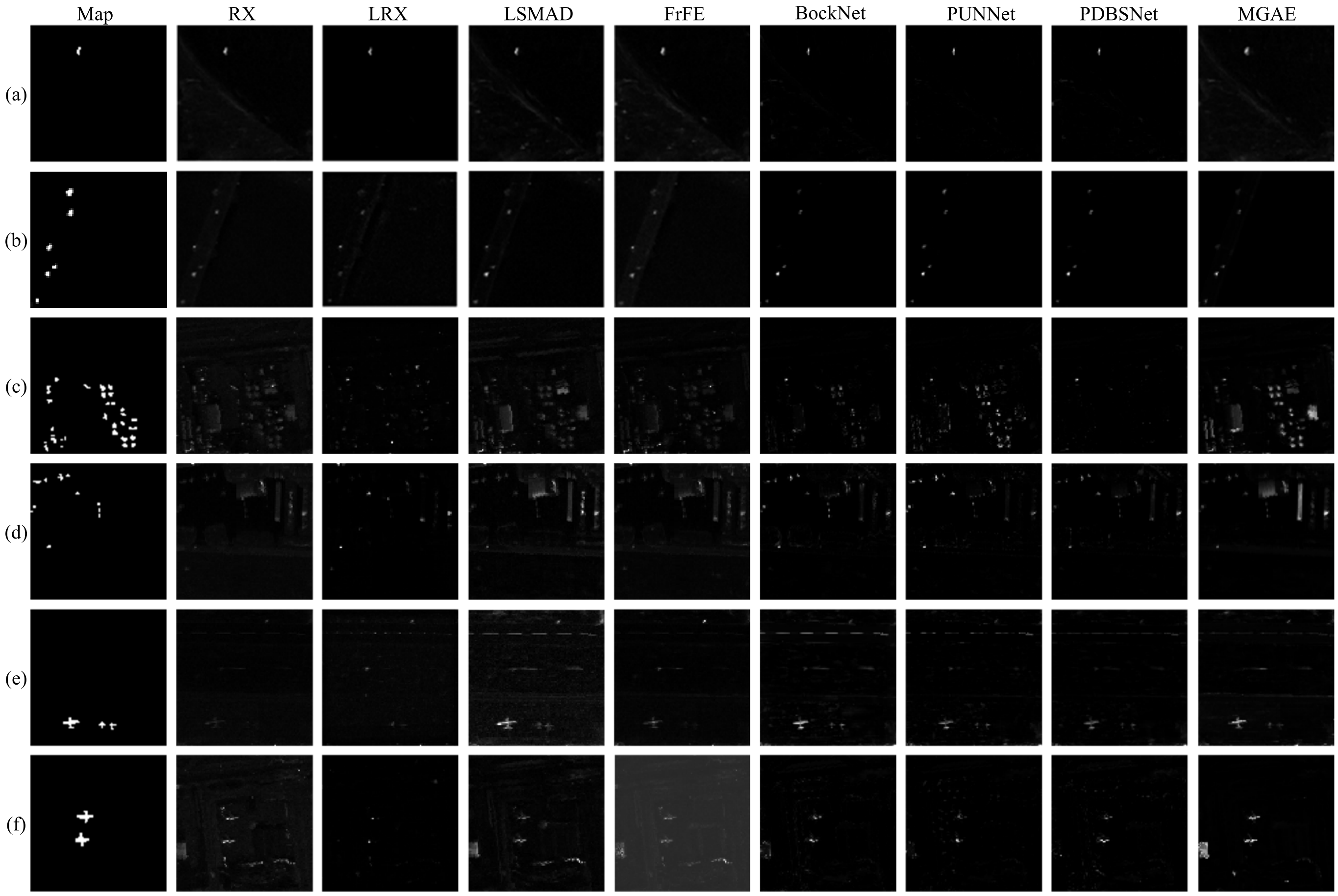

Fig. 8. Label maps of the selected datasets and anomaly maps of all methods. (a) BEACH-I. (b) BEACH-II. (c) HYDICE. (d) Gainesville. (e) Gulfport. (f) Los Angeles.

confirming robustness in challenging environments. Collectively, these curves demonstrate MGAE's high detection accuracy and scene adaptability across an extended threshold range, significantly outperforming other algorithms.

2) 3D_ROC: The Fig. 10 presents the 3D_ROC performance of eight anomaly detection algorithms across six hyperspectral datasets, jointly evaluating false positive rate (FPR), decision threshold (τ), and true positive rate (TPR) to characterize detection trade-offs. MGAE demonstrates significant advantages across all scenarios: on homogeneous backgrounds (e.g., BEACH-I) with FPR < 0.3 and $\tau \in [0.2, 0.6]$, it sustains TPR > 0.9, surpassing RX (TPR < 0.7) and LRX (TPR < 0.6) by $30 - 50$ percentage points. Under complex environments like HYDICE (urban heterogeneity) and Los Angeles (spectral confusion), MGAE achieves higher TPR at equivalent FPR through relaxed thresholds (τ reduced by $0.1–0.2$), significantly outperforming deep learning counterparts (BockNet, PDBSNet). This superiority stems from MGAE's graph-structural hyperspectral modeling: graph attention mechanisms capture spectral-spatial correlations while dual-mask loss suppresses background interference, maintaining dynamic high-detection/low-FPR balance across wide threshold ranges. In contrast, traditional methods (e.g., RX, limited by Gaussian assumptions) and deep approaches (e.g., PUNNet, prone to local overfitting) exhibit flatter curves requiring strict thresholds ($\tau < 0.2$) or high FPR tolerance (> 0.4) to approach MGAE's performance, validating its robustness in complex hyperspectral scenarios.

3) Box Plot: The Fig. 11 displays boxplots comparing anomaly detection scores (green) and background scores (red) across methods, revealing MGAE's consistent superiority. In BEACH-I, it achieves near-maximal anomaly scores with minimal background interference, effectively handling spectral variability. While experiencing a slight decline in BEACH-II, it maintains competitive robustness across beach environments. MGAE dominates in HYDICE (complex urban areas) and Gainesville, demonstrating cross-dataset stability, and excels in Gulfport's airport scene with optimal anomaly-background separation. For the spectrally confounded Los Angeles dataset, MGAE sustains exceptional anomaly detection with suppressed background. Crucially, MGAE markedly outperforms traditional methods (RX/LRX/LSMAD) in challenging environments featuring diverse anomalies, while its sustained low background scores confirm precise spectral discrimination—addressing hyperspectral imaging's fundamental challenge of anomaly-background separation.

4) Summary: As evidenced by Fig. 8 and Table I, MGAE achieves superior AUC values across all datasets, consistently

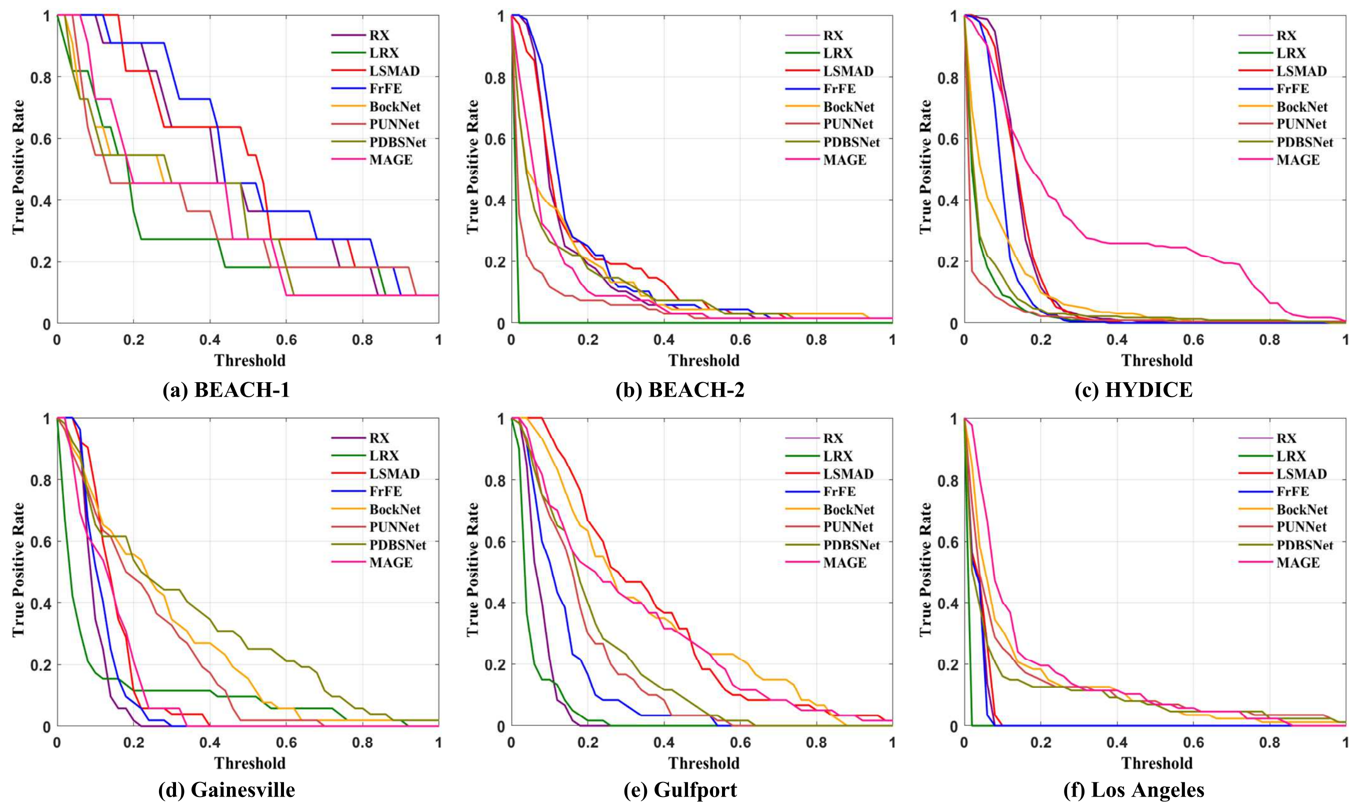

Fig. 9. 2D_ROC curves of all methods for the six datasets.

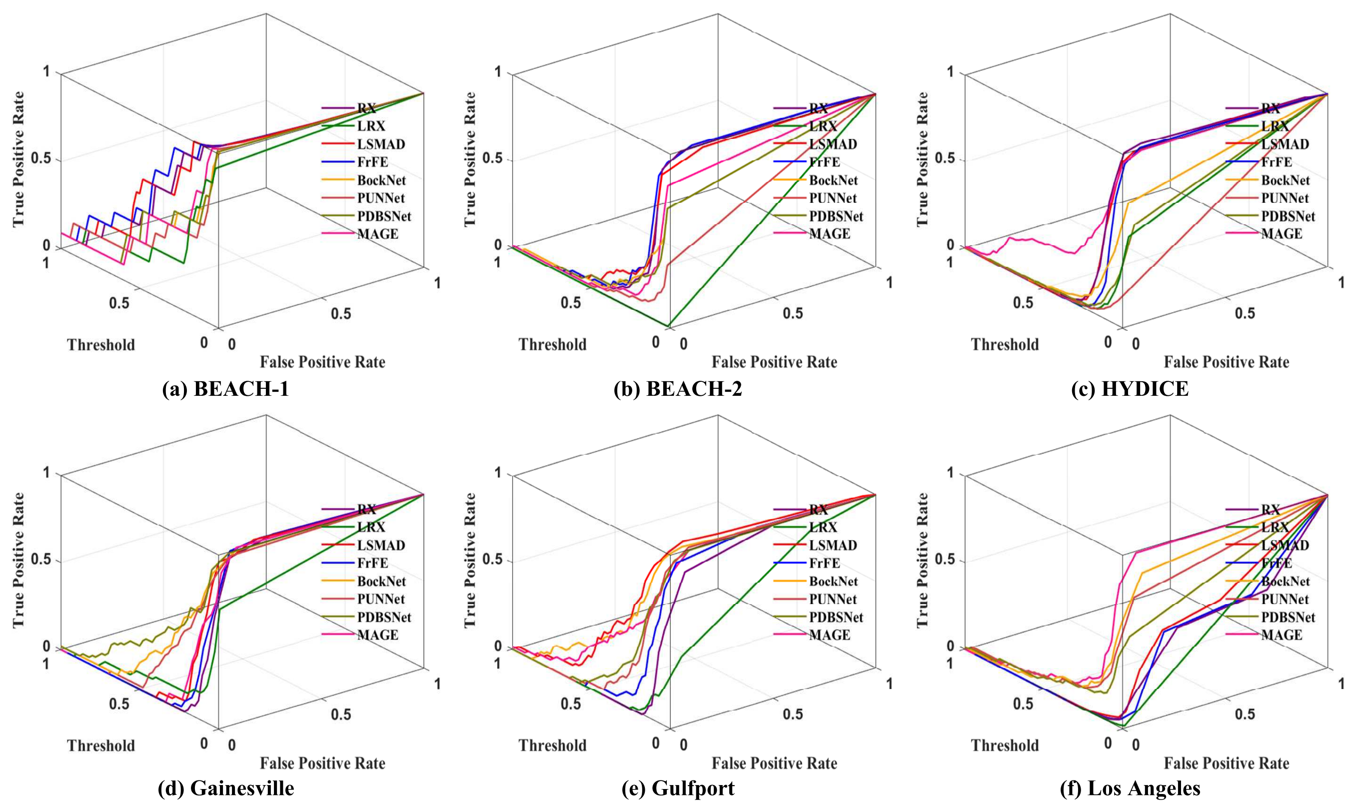

Fig. 10. 3D_ROC curves of all methods for the six datasets.

outperforming comparative methods. Corroborated by ROC characteristics, MGAE maintains higher true positive rates and lower false positive rates than traditional approaches (e.g., RX, LSMAD) in every operational environment. This demonstrates

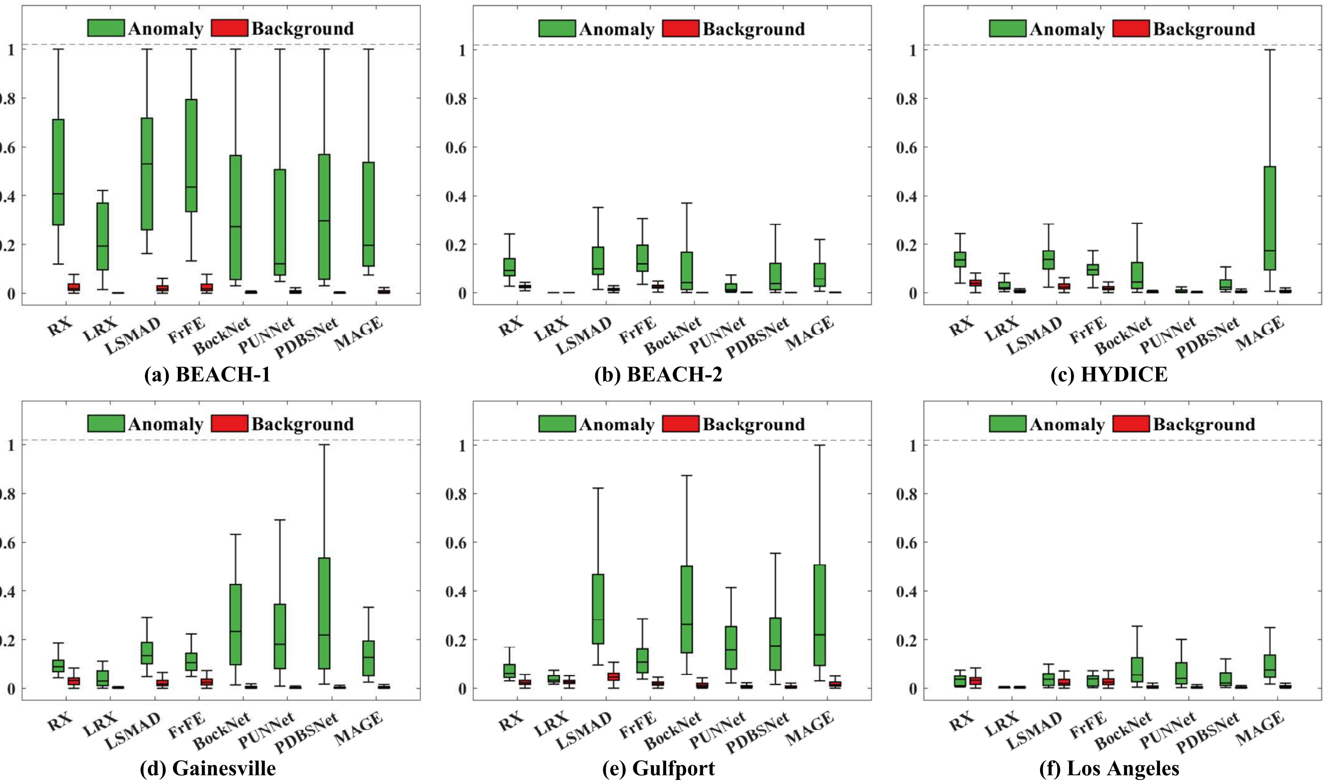

Fig. 11. Box plots of all methods for the six datasets.

TABLE II
AUC VALUES OF ABLATION EXPERIMENTS ON SIX DATASETS

GAT	Re-masking strategy	Twice Loss	BEACH-I	BEACH-II	HYDICE	Gainesville	Gulfport	Los Angeles
✓	×	×	0.9316	0.9713	0.9190	0.7691	0.6791	0.8914
✓	✓	×	0.9871	0.9832	0.9628	0.9719	0.9742	0.9773
✓	×	✓	0.9913	0.9829	0.9651	0.9792	0.9795	0.9788
✓	✓	✓	0.9999	0.9946	0.9741	0.9913	0.9923	0.9833

precise anomaly discrimination with minimal false alarms — a critical advantage for real-world deployment. The 3D performance evolution further confirms MGAE's stability and adaptability across threshold configurations. Collectively, these cross-dataset analyses validate MGAE's robust efficacy for HAD.

D. Ablation Study

This section analyzes the effectiveness of the proposed re-masking strategy and twice loss strategy. Using a traditional Graph Attention Network (GAT) as the baseline framework, the proposed MGAE incorporates an innovative re-masking strategy that mitigates decoder-induced reconstruction error averaging by preventing simplistic mapping relationships. Concurrently, a twice loss strategy is introduced to balance background modeling accuracy and anomaly detection sensitivity. Table II compares the AUC performance of the three models across six datasets. The re-masking strategy alone provides modest improvements over the baseline GAT, as background information loss during re-masking

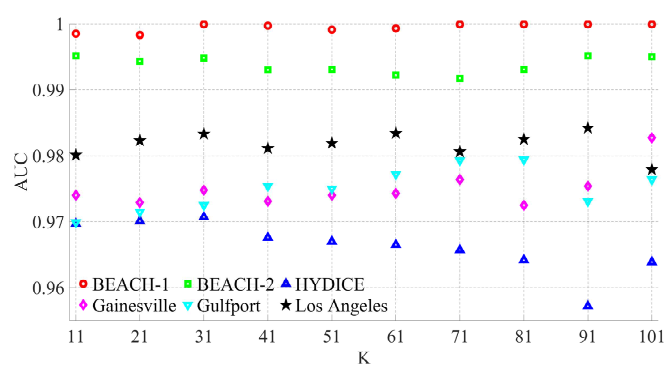

Fig. 12. The impact of parameter k selection on the AUC values changes in the results of the experimental dataset.

can introduce false anomalies, while random masking uncertainty may reconstruct anomalies—increasing false positive rates. The twice loss strategy addresses these limitations by preserving global background context while minimizing constraints on anomalies to suppress their reconstruction.

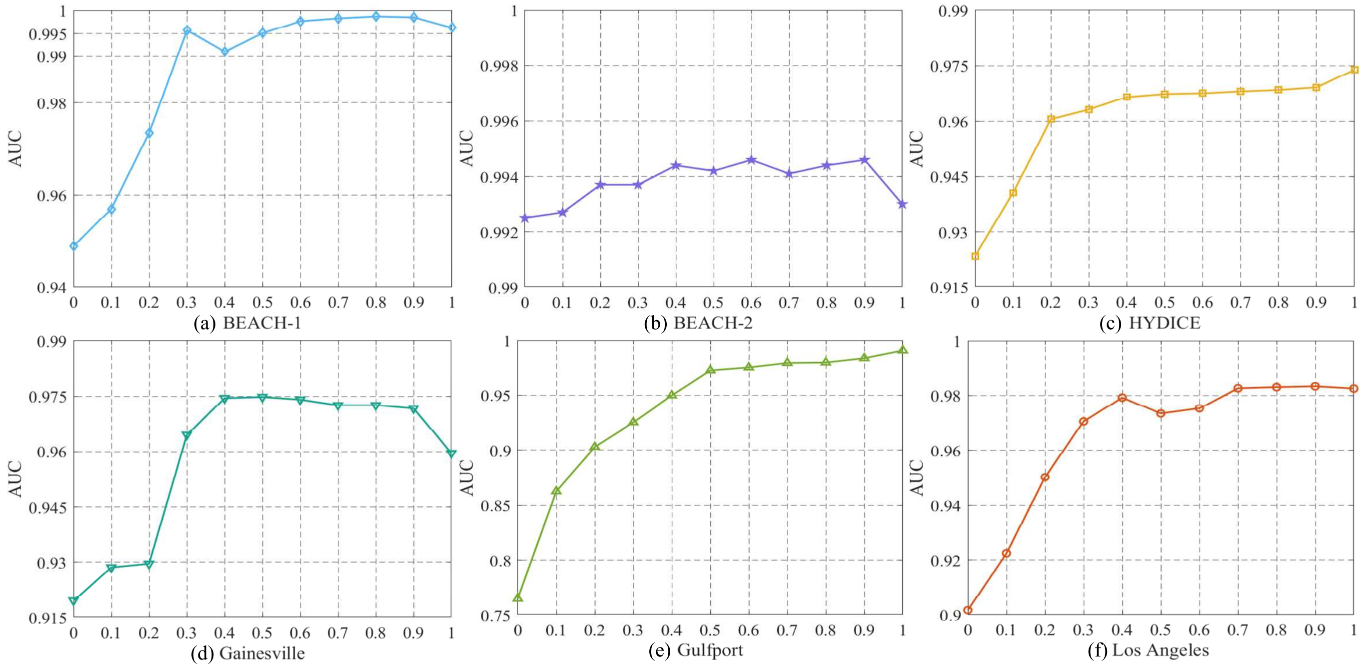

Fig. 13. The variation curve of AUC values in the experimental dataset results caused by different parameters α.

Consequently, MGAE integrating both strategies achieves significant enhancement in HAD capability.

E. Parameter Analysis

In this section, we analyze the influence of the parameters k and α on the detection performance of the proposed MGAE method.

1) Parameter k: The parameter k is critical in the MGAE method as it defines the number of nearest neighbors considered during the anomaly detection process. This parameter's influence on the detection performance is profound, as it directly affects both the sensitivity and specificity of the method. As shown in Fig. 12, our analysis of six datasets with varying k-values from 11 to 101 has demonstrated the critical role of k in the performance of the MGAE method. The sensitivity and specificity of anomaly detection are significantly influenced by the choice of k, affecting the overall accuracy and reliability of the method.To determine the optimal k value, we analyze the AUC values for different k-values on each dataset. The data revealed that datasets such as BEACH-I consistently performed well across a broad range of k-values (maintaining AUC between 0.99 and 0.98), indicating robustness to the choice of k. In contrast, datasets like Gainesville exhibited substantial fluctuations in performance (AUC rising from 0.96 to 0.98), highlighting a sensitivity to k-value selection. For instance, the Gainesville dataset showed improved performance as k increased from 11 to 101, peaking at $k = 101$ with the highest AUC value of 0.98. This suggests that a higher k-value may be necessary to capture the broader context within this particular dataset, reducing noise and minimizing false positives. Complex urban environments (HYDICE, Los Angeles) require $k \geq 71$ to stabilize performance (> 0.97 AUC) by suppressing outlier-induced false positives, whereas structured scenes (Gulfport) achieve peak discrimination at moderate k-values (51-71, AUC 0.97) through balanced spatial-spectral modeling. This demonstrates that optimal k is intrinsically linked to scene heterogeneity.

2) Parameter α: The parameter α controls the proportion of reconstruction loss and smoothness loss in our proposed Twice Loss. In order to make the effect of parameter A more intuitive, we set the value of k in the parameter experiment. Fig. 13 illustrates the impact of parameter α on the Area Under the Receiver Operating Characteristic (ROC) curve (AUC) values across different datasets.

The Fig. 13 and Table III demonstrate the critical influence of the loss-balancing parameter α on MGAE's AUC performance across hyperspectral datasets. For beach environments, BEACH-I exhibits monotonic AUC improvement with increasing α, stabilizing near $\alpha = 0.3$ and peaking at 0.9987 ($\alpha = 0.8$); BEACH-II maintains stable performance despite minor fluctuations at $\alpha = 0.2$, achieving dual maxima of 0.9946 at $\alpha = 0.6$ and 0.9. Urban and mixed scenes show distinct patterns: Gainesville displays rapid AUC escalation beyond $\alpha = 0.3$, stabilizing at its peak 0.9748 ($\alpha = 0.5$), while Los Angeles manifests gradual improvement culminating at maximum AUC ($\alpha = 0.9$). Crucially, when $\alpha = 1$ (pure reconstruction loss without smoothness regularization), HYDICE and Gulfport attain their highest AUC values - indicating smoothness loss provides negligible benefit for these specific environments. These findings establish α's dual role in HAD: 1) Optimizing the reconstruction-smoothness tradeoff enhances model efficiency, and 2) Enabling scene-specific performance customization through spectral-characteristic-adaptive loss balancing. This tradeoff underscores α's critical function in tailoring detection precision to diverse operational scenarios while maintaining computational effectiveness.

TABLE III

THE SELECTION OF PARAMETER α AND THE IMPACT OF DIFFERENT α VALUES ON THE AUC OF EXPERIMENTAL RESULTS ACROSS SIX DATASETS

alpha	BEACH-1	BEACH-2	HYDICE	Gainesville	Gulfport	Los Angeles
0.0	0.9489	0.9925	0.9235	0.9196	0.7652	0.9017
0.1	0.9570	0.9927	0.9406	0.9286	0.8627	0.9224
0.2	0.9735	0.9937	0.9606	0.9296	0.9030	0.9503
0.3	0.9957	0.9937	0.9632	0.9645	0.9254	0.9706
0.4	0.9911	0.9944	0.9666	0.9745	0.9499	0.9735
0.5	0.9951	0.9942	0.9674	**0.9748**	0.9727	0.9754
0.6	0.9977	**0.9946**	0.9676	0.9741	0.9754	0.9794
0.7	0.9983	0.9941	0.9681	0.9726	0.9795	0.9826
0.8	**0.9987**	0.9944	0.9685	0.9726	0.9799	0.9830
0.9	0.9985	**0.9946**	0.9692	0.9717	0.9838	**0.9833**
1.0	0.9963	0.9930	**0.9739**	0.9596	**0.9909**	0.9825

TABLE IV

INFERENCE TIME OF DIFFERENT DETECTION METHODS ON SIX DATASETS

Dataset	RX	LRX	LAMAD	FrFE	BockNet	PUNNet	PDBSNet	MGAE
BEACH-I	3.7479	27.7776	2.2543	57.5919	0.20	1.6327	0.2137	**0.0296**
BEACH-II	2.0345	11.5941	1.4808	57.7422	0.19	1.6566	0.1878	**0.0285**
HYDICE	2.0693	34.1235	2.1119	62.6623	0.18	1.6348	0.2065	**0.0438**
Gainesville	2.4934	31.1393	2.1681	67.8051	0.19	1.5893	0.2118	**0.0212**
Gulfport	2.5542	32.1954	2.2015	67.2539	0.21	1.5754	0.2093	**0.0336**
Los Angeles	2.5353	37.6643	2.2184	68.2960	0.20	1.5628	0.2132	**0.0311**

F. Comparison of Inference Times

The Table IV reports the inference times of various comparison methods across six datasets. Except for BockNet, PUNNet, PDBSNet, and our MGAE (implemented using the PyTorch framework), all other methods were coded in MATLAB 2021b. The five traditional physics-driven methods (RX, LRX, LAMAD, FrFE) exhibit significantly longer inference times across most datasets compared to their deep learning-based counterparts. Although MGAE involves graph structure construction and graph attention-based feature aggregation — introducing marginal computational overhead — it consistently achieves the shortest inference time across all datasets, notably outperforming BockNet and PDBSNet by an order of magnitude in efficiency. This result underscores MGAE's superior balance between model complexity and computational efficiency.

V. CONCLUSION

In this article, we introduce Masked Graph Autoencoder (MGAE), a pioneering approach designed for HAD. This innovative method leverages graph neural networks (GNN) to learn graph representations unsupervised. At the core of MGAE's strategy is the reconstruction of masked node features, which is a significant departure from traditional approaches. Unlike conventional Graph Autoencoders (GAE) that utilize multilayer perceptrons (MLP) for decoding, we implement a re-mask decoding strategy using GNN to enhance the capability of MGAE. Additionally, within the GNN framework, we integrate the Laplacian matrix with the reconstructed feature matrix. This integration is crucial as it significantly boosts the model's sensitivity to topological variations and irregularities within the data structure. Such an approach allows for the effective capture of nuanced anomaly patterns by leveraging the inherent graph structure encoded by the Laplacian matrix, thereby augmenting the detection capabilities of our model. By incorporating these advanced graph-based insights, MGAE demonstrates an ability to outperform conventional methods, particularly in detecting subtle and complex anomalies in hyperspectral images. This advancement establishes MGAE as a potent tool in the hyperspectral imaging field, providing enhanced accuracy and reliability in anomaly detection.

The experimental results on six real HSIs demonstrate that our approach outperforms other state-of-the-art HAD methods. Future work will focus on integrating spatial information into the MGAE framework to enhance anomaly detection by leveraging both spectral and spatial features. Additionally, optimizing model parameters adaptively and exploring more refined weight definitions between graph nodes will be pursued to improve sensitivity to subtle anomalies. Finally, incorporating advanced learning techniques like self-supervised learning could further boost generalization across diverse hyperspectral datasets, pushing the boundaries of detection accuracy.

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Bing Tu (Senior Member, IEEE) received the M.S. degree in control science and engineering from Guilin University of Technology, Guilin, China, in 2009, and the Ph.D. degree in mechatronic engineering from Beijing University of Technology, Beijing, China, in 2013.

From 2015 to 2016, he was a Visiting Researcher with the Department of Computer Science and Engineering, University of Nevada, Reno, NV, USA, which was supported by China Scholarship Council. He is currently a Full Professor with Nanjing University of Information Science and Technology, Nanjing, China. His research interests include sparse representation, pattern recognition, and analysis in remote sensing. He is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

Baoliang He (Student Member, IEEE) received the B.S. degree in electrical engineering and automation from Henan Polytechnic University, Jiaozuo, Henan, China, in 2025. He is currently pursuing the M.S. degree in optical engineering with Nanjing University of Information Science and Technology, Nanjing, Jiangsu, China. His research interests include hyperspectral anomaly detection.

Yan He (Member, IEEE) received the Ph.D. degree in computer science from Hunan University, Changsha, Hunan, China, in 2025. She is currently a Lecturer with Nanjing University of Information Science and Technology, Nanjing, China. Her research interests include hyperspectral image processing and remote sensing image registration.

Tao Zhou (Student Member, IEEE) is currently pursuing the master's degree in electronic science and technology with the School of Electronic and Information Engineering, Anhui Jianzhu University, Hefei, Anhui, China.

Bo Liu (Member, IEEE) received the B.S. and Ph.D. degrees in optical engineering from Beijing University of Posts and Telecommunications, Beijing, China, in 2008 and 2013, respectively. He is currently a Professor with the School of Physics and Optoelectronics, NUIST, China. His research interests include all-optical signal processing, radio over fiber, and broadband optical communication.

Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunica ções, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

From 2013 to 2021, she was a Full Professor with Sun Yat-sen University. Since 2022, she has been with China University of Geosciences as a Full Professor. She has received several prestigious funding grants at the national and international levels. She has authored more than 160 Journal Citation Report (JCR) articles, 60 international conference papers, and a book chapter.

Dr. Li has been serving as the Editor-in-Chief for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING since 2021.

Antonio Plaza (Fellow, IEEE) received the M.Sc. and Ph.D. degrees in computer engineering from the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, Cáceres, Spain, in 1999 and 2002, respectively.

He is currently the Head of the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura. He has authored more than 600 publications, including over 200 JCR journal articles (over 160 in IEEE journals), 23 book chapters, and around 300 peer-reviewed conference proceeding papers. His research interests include hyperspectral data processing and parallel computing of remote sensing data.

Dr. Plaza was an Editorial Board Member of the IEEE Geoscience and Remote Sensing Newsletter from 2011 to 2012 and *IEEE Geoscience and Remote Sensing Magazine* in 2013. He was also a member of the Steering Committee of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. He is a fellow of IEEE for contributions to hyperspectral data processing and parallel computing of Earth observation data. He was a recipient of the recognition of Best Reviewers of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS in 2009 and the Best Reviewer of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2010, for which he has served as an Associate Editor from 2007 to 2012. He was also a recipient of the Best Column Award of *IEEE Signal Processing Magazine* in 2015, the 2013 Best Paper Award of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the Most Highly Cited Paper (2005–2010) in *Journal of Parallel and Distributed Computing*. He received best paper awards at the IEEE International Conference on Space Technology and the IEEE Symposium on Signal Processing and Information Technology. He has served as the Director of Education Activities for the IEEE Geoscience and Remote Sensing Society (GRSS) from 2011 to 2012 and the President of Spanish Chapter of IEEE GRSS from 2012 to 2016. He has reviewed more than 500 manuscripts for over 50 different journals. He has served as the Editor-in-Chief for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2013 to 2017. He has guest-edited ten special issues on hyperspectral remote sensing for different journals. He is an Associate Editor of the IEEE ACCESS (receiving the recognition as an Outstanding Associate Editor of the journal in 2017). Additional information is available at http://www.umbc.edu/rssipl/people/aplaza